

Anesthetic approaches and 2-year recurrence rates in non-muscle invasive bladder cancer: a randomized clinical trial.

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The effect of anesthesia methods on non-muscle invasive bladder cancer (NMIBC) recurrence post-resection remains uncertain. We aimed to compare the oncological outcomes of spinal anesthesia (SA) and general anesthesia (GA) in patients with NMIBC. This prospective randomized controlled trial recruited 287 patients with clinical NMIBC at Seoul National University Hospital from 2018 to 2020. The patients underwent transurethral resection of the bladder tumor within 4 weeks of randomization. Intrathecal hyperbaric bupivacaine (0.5%) and a mixture of propofol (1-2 mg/kg) and fentanyl (50-100 µg/kg) were used as induction agents in the SA and GA groups, respectively, with desflurane or sevoflurane used for maintaining anesthesia. The primary and secondary outcome measures were disease recurrence and disease progression, respectively, at 2 years after resection. Cumulative incidence of outcomes was compared between the two groups using time-to-event analyses. 15 patients required alternative anesthesia owing to clinical needs such as SA failure or significant obturator reflex, resulting in a modified intention-to-treat (ITT) population of 272 patients. Time-to-event analysis showed a significantly lower recurrence of NMIBC in the SA group than in the GA group, in both ITT (27.4% vs 39.8%) and modified ITT populations (26.8% vs 39.6%). Disease progression occurred more frequently in the GA than in the SA group (15.2% vs 7.8%), although the difference was not statistically significant. A notable reduction in the 2-year recurrence rate was observed in patients who underwent SA than in those who underwent GA. Thus, SA may be considered the preferred anesthetic approach. NCT03597087.

Genetic instability of a single exposure to sevoflurane at different concentrations in monitored mice.

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Sevoflurane is an inhalation anesthetic widely used for general anesthesia, but its genotoxic potential is controversial in clinical studies. It is unknown whether the effects are due to surgery or the anesthetic. Thus, for the first time, the present study investigated genotoxicity in peripheral blood cells and in target organs (liver, lung, and kidney) and micronucleus (MN) in the bone marrow of a single exposure to sevoflurane at three different concentrations in monitored mice. Ninety Swiss mice were distributed into the following groups: exposure to sevoflurane at 3.3% (low), 4.5% (intermediate), and 6.0% (high) in 40% oxygen (O₂) for 2 h; negative control (no exposure); negative control with O₂; and positive control. The exposed animals were heated, monitored for vital signs (temperature, O₂ saturation, heart rate/pulse, and respiratory rate), and anesthetized via a modern low-flow digital system. Mice were euthanized 2 and 24 h after exposure for evaluation by the comet assay and MN test, respectively. No DNA damage occurred in the 3.3% group for any of the organs evaluated, and no genotoxic or mutagenic effects were observed at any sevoflurane concentration in the peripheral blood or liver cells. However, a significant increase in DNA damage was observed at higher concentrations in kidney (4.5%) and lung cells (6.0%) and in the MN frequency (groups 4.5% and 6.0%). No cytotoxicity or histological alterations were observed. In conclusion, high concentrations of sevoflurane induce DNA damage, but concentrations equivalent to those used in clinical practice do not demonstrate genotoxic or mutagenic effects.

New inhaled anesthetics.

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Desflurane and sevoflurane provide one clear advantage over other currently available potent inhaled anesthetics. Their lower solubilities permit a more precise control over the delivery of anesthesia and a more rapid recovery from anesthesia. Most of their other properties reflect similar properties of their predecessors--with a few exceptions. Indeed, at concentrations of 1 MAC or less, the pharmacologic properties of these two agents differ little if at all. However, in contrast to desflurane, at concentrations exceeding 1 MAC sevoflurane has little or no respiratory irritant properties and may be used to rapidly induce anesthesia. Neither anesthetic seems to materially affect heart rate at concentrations lower than MAC, but at higher concentrations desflurane, but not sevoflurane, may increase heart rate. Desflurane strongly resists biodegradation and degradation by soda lime, whereas sevoflurane is vulnerable to degradation and the degradation by soda lime or Baralyme produces a toxic product. Thus, neither of these new anesthetics can be said to be ideal, but each may be a step in that direction.

Metabolism and toxicity of the new anesthetic agents.

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Human biotransformation of the newer anesthetics, desflurane and sevoflurane, has been characterized in vitro and in vivo. Major metabolites of desflurane are inorganic fluoride and trifluoroacetyl chloride, which may bind to tissue proteins or form trifluoroacetic acid, which is excreted in urine. Major metabolites of sevoflurane are fluoride and hexafluoroisopropanolol (HFIP), which is rapidly glucuronidated and excreted as HFIP-glucuronide in urine. The rate of sevoflurane metabolism is approximately one-third that of methoxyflurane, and 1.5-2 times that of enflurane, while that of desflurane is minimal. The extent of metabolism of sevoflurane and desflurane is 2-5% and 0.02-0.2% of the dose taken up, respectively. Peak serum fluoride concentrations occur within one hour after sevoflurane anesthesia, are usually in the range of 20-40 microM, and decline rapidly. Fluoride concentrations have exceeded 50 microM in approximately 7% of sevoflurane patients. For sevoflurane, the plasma fluoride concentration-time profile most closely resembles that of enflurane. For desflurane, elevations of plasma metabolite concentrations can only be detected after prolonged anesthesia. Investigations to date suggest that the biotransformation of sevoflurane does not appear to result in clinically significant hepatotoxicity or nephrotoxicity. Only a single case of immune-mediated desflurane hepatotoxicity has been reported.

Neurocirculatory responses to sevoflurane in humans. A comparison to desflurane.

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Sevoflurane and desflurane are new volatile anesthetics with low blood solubilities that confer properties of rapid anesthetic induction and emergence. Desflurane has been associated with neurocirculatory excitation after the rapid increase in inspired concentrations. The current study evaluated and compared the sympathetic and hemodynamic responses associated with the administration of sevoflurane to those associated with administration of desflurane in humans. After Institutional Review Board approval, 21 healthy, young (19-32 yr) volunteers were randomly selected for participation. Arterial and central

venous pressures were measured directly, and heart rate, forearm blood flow, and plasma norepinephrine concentrations were determined indirectly. Efferent muscle sympathetic nerve activity was recorded by microneurography. After neurocirculatory recordings at conscious baseline, measurements were repeated beginning 2 min after 2 mg/kg propofol while the anesthetic was increased incrementally by mask over a 10-min period at 1%, 2%, and 3% sevoflurane ($n = 12$) or 3%, 6%, and 9% desflurane ($n = 9$). Responses to intubation were recorded and, 20 min later, recordings were evaluated during steady-state periods of 0.41, 0.83, and 1.24 MAC. Data also were obtained after steady-state 0.83 MAC measurements when the inspired gas concentration was rapidly increased to either 3% sevoflurane or 9% desflurane ("transition" to 1.24 MAC). Neurocirculatory variables did not differ between the two groups at conscious baseline. During the period of administration via mask and during the "transition" period, the significant increases in sympathetic nerve activity, heart rate, mean arterial pressure, and central venous pressure associated with desflurane were not observed with sevoflurane. Ten minutes after induction, mean arterial pressure and heart rate responses to intubation did not differ between groups. With increasing anesthetic concentration, there were progressive and similar decreases in mean arterial pressure in both groups and no changes in heart rate. Central venous pressure, sympathetic nerve activity, and plasma norepinephrine increased with the greater minimum alveolar concentration multiple of desflurane but not with that of sevoflurane. The neurocirculatory excitation seen with rapid increases in desflurane did not occur with sevoflurane. At steady-state, increasing the concentration of sevoflurane was associated with lower sympathetic nerve activity and central venous pressure and similar mean arterial pressure and heart rate with that of desflurane.

[New inhalation anesthetics].

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Recently, two new halogenated volatile anaesthetics, sevoflurane and desflurane, have been approved for clinical use in Germany. Their low solubility in blood is the most important common property, and this represents the most obvious difference from the inhalational anaesthetics currently used. Extensive clinical and experimental evaluations have confirmed the superior pharmacokinetic properties predicted. Both sevoflurane and desflurane provide more rapid emergence from anaesthesia, permit easier titration of the anaesthetic dose during maintenance and offer more rapid recovery from anaesthesia. For sevoflurane, there are additional advantages: a pleasant odor, negligible airway irritation, and excellent pharmacodynamic characteristics that even provide cardiovascular stability comparable to isoflurane. A certain disadvantage and source of potential nephrotoxicity result from the metabolism of sevoflurane (2-5%) to anorganic fluoride and degradation to compound A in carbon dioxide absorbents. The extensive clinical data reported to date have revealed no evidence that sevoflurane has adverse renal effects. New insight into the pathomechanism of nephrotoxicity associated with either production of fluoride or compound A may well support clinical experience. Desflurane strongly resists *in vivo* metabolism and because of this it appears to be devoid of toxicity. Nevertheless, potential side-effects may result from degradation in dry absorbents and subsequent release of CO, from its extreme pungency and irritating airway effects. Thus, desflurane is not recommended for induction of anaesthesia, especially in children. The tendency for desflurane transiently to stimulate sympathetic activity, especially at concentrations above 1.0 MAC, limits its application in patients with cardiac disease.

New drugs in anesthesia.

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The lower solubilities of desflurane and sevoflurane provide a clear advantage over older potent agents, permitting more precise control over the delivery of anesthesia and more rapid recovery from anesthesia. Other pharmacological properties of these two agents at concentrations of 1 MAC or less differ little from those of older agents. Sevoflurane differs from desflurane in that it is without pungency and useful for induction of anesthesia by inhalation. It is less prone to increase pulse rate at higher concentrations. Desflurane also can cause transient increases in pulse rate and blood pressures when its concentration is rapidly increased above 6%. Sevoflurane has the disadvantage of vulnerability to degradation in vitro and in vivo, raising a concern regarding a subtle toxicity. Although neither of these new anesthetics can be said to be ideal, each may be a step in that direction.

The role of new anesthetic agents.

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The three anesthetic drugs introduced most recently to the market are sevoflurane, desflurane, and ropivacaine. Sevoflurane and desflurane are both inhalational anesthetic agents and ropivacaine is a local anesthetic agent. Sevoflurane provides a rapid onset and offset of action; it is well tolerated with little airway irritation. It is hemodynamically stable, with low potential for toxicity. Concerns about its interaction with soda lime during low-flow anesthesia with the production of Compound A have not proved to be a clinical problem. While desflurane also provides rapid onset and recovery from anesthesia, it is not as hemodynamically stable as sevoflurane, and also causes airway irritation. Ropivacaine is a unique local anesthetic in that it is supplied as the pure S-enantiomer. It is at least as effective as bupivacaine, with lower toxicity and less motor block for the same degree of sensory block.

[A role for sevoflurane in ambulatory surgery].

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Sevoflurane is a viable alternative to propofol and desflurane for both induction and maintenance of general anaesthesia in the ambulatory setting. As a result of sevoflurane's lack of respiratory irritant properties, it provides for a smooth induction and prompt emergence from outpatient anaesthesia. In addition, the relatively low incidence of post-operative nausea and vomiting facilitates "fast-tracking" after ambulatory surgery. Although no single anaesthetic agent is ideal, when sevoflurane is combined with other adjunctive drugs it can produce excellent surgical conditions for a wide variety of ambulatory surgical and diagnostic procedures.

[Volatile anesthetics].

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None of the currently available inhaled anesthetics has all of the properties of an "ideal" inhaled agent. The exceptionally low solubility of desflurane and sevoflurane offers a significantly greater precision of

control over maintenance of anesthesia and a potential for a more rapid recovery from anesthesia than other inhaled anesthetics. Sevoflurane appears to offer some advantages regarding cardiovascular stability. Products of metabolism or degradation can be associated with potential organ-specific toxic effects. Renal toxicity is discussed for enflurane and sevoflurane. Breakdown products of volatile agents with carbon dioxide absorbents have to be mentioned especially for sevoflurane (compound A) and desflurane (CO). In contrast to intravenous anesthetics, volatile anesthetics are associated with cardio- and cerebroprotection.
